

Qualifying Exam in Microeconomics
Arizona State University
June 2015

There are four questions and you have four hours to complete the exam.

1. There are L goods. Consider an agent whose consumption set is the non-negative orthant $\mathbb{R}_+^L$. The agent's preferences, denoted by $\succeq$, are locally non-satiated, complete, and transitive. The agent's endowment e is in his or her consumption set. At a price p^* , the bundle x^* belongs to the agent's demand set.

- ~~(a)~~ Please prove that $x' \succ x^* \implies p^* \cdot x' > p^* \cdot x^*$. *check the notes!*
~~(b)~~ Please prove that $x' \succeq x^* \implies p^* \cdot x \geq p^* \cdot x^*$.

2. Consider an economy with a single physical good and two states of the world. Consider two consumers in this economy, a and b , each with consumption set $\mathbb{R}_+^2$. For $i = a, b$, consumer i has subjective expected utility preferences with belief $p_s \in (0, 1)$ that the state is $s \in 1, 2$, and (state-independent) vN-M utility u_i on $\mathbb{R}_+$. Each consumer is endowed with consumption of w in each state of the world (henceforth the consumer's *wealth*).

Recall the definition of a "more concave" function: For two continuous, strictly-increasing, real-valued functions f and g on $\mathbb{R}_+$, g is *more concave than* f if there is a *concave* function $T : \text{Range}(f) \rightarrow \mathbb{R}$ with $g(z) = T(f(z))$ for every $z \geq 0$. (Of course T is strictly increasing since f and g are.)

Assume throughout that u_i is *continuous* and *strictly increasing* for $i = 1, 2$.

Finally for $i = a, b$ define for each $w > 0$,

$$A_i(w) = \{(x_1, x_2) \in \mathbb{R}^2 \mid (w + x_1, w + x_2) \geq 0 \text{ and } E[u_i(w + x)] \geq u_i(w)\}$$

where, in an abuse of notation, $E[u_i(w + x)] = p_1 u_i(w + x_1) + p_2 u_i(w + x_2)$. The set $A_i(w)$ is consumer i 's *acceptance set* at sure wealth w , the set of risky net trades from w that consumer i weakly prefers to consuming a sure wealth of w .

- ~~(a)~~ Prove that if u_b is more concave than u_a , then $A_b(w) \subseteq A_a(w)$ for every $w > 0$.
~~(b)~~ Recall that consumer i 's *certainty equivalent* for $(w + x_1, w + x_2) \geq 0$ is the sure wealth $CE_i(x, w)$ that is indifferent to $(w + x_1, w + x_2)$: $u_i(CE_i(x, w)) = E[u_i(w + x)]$. Prove that if $A_b(w) \subseteq A_a(w)$ for every $w > 0$, then $CE_b(x, w) \leq CE_a(x, w)$ for every $w > 0$ and $x \geq -(w, w)$.
~~(c)~~ For this part assume as a simplification that each u_i is C^2 . Prove that if $A_b(w) \subseteq A_a(w)$ for every $w > 0$, then u_b is more concave than u_a .

3. The owner of a restaurant hires an individual to manage the restaurant. The manager can choose either high effort or low effort ($e \in \{e_h, e_l\}$). The cost to the manager of choosing high effort is $c > 0$ and the cost of low effort is zero. When the manager chooses high effort, the restaurant generates a revenue r_h with probability $0 < p < 1$, and revenue r_l otherwise. When the manager chooses low effort, the revenue is zero with probability one. The manager's reservation utility is zero. The manager's utility when paid a wage w and choosing high effort is $w^{0.5} - c$. Finally, assume that $r_h - r_l > \frac{c^2}{p^2}$.

(a) What contract should be offered by the owner? *→ he doesn't observe r !*

(b) The situation is modified. Suppose the owner has the option to monitor the manager by appearing at the restaurant by surprise. If the manager chose e_h , there is a probability $0 < q < 1$ that the owner will observe it. If the manager chose e_l , the owner observes nothing with probability one. Suppose also that a new law prohibits contracts based on revenue, that is to say, the owner can offer differential wages based only on whether or not the owner observes that the manager chose e_h .

What contract should be offered by the owner?

(c) Suppose the owner wishes to implement e_h in both problems, 3a and 3b. Under what conditions is the owner better off in situation 3b than in situation 3a.

$u(e_l) = w$?

4. Consider a Cournot duopoly in which the firms sell identical products and have the same linear cost function with unit cost of $c > 0$. The inverse demand $P(q)$ is continuous on $\mathbb{R}_+$ and there is an output $\bar{q} > 0$ such that: $0 \leq P(q) < c$ for every $q \geq \bar{q}$; and on $[0, \bar{q}]$, $P(\cdot)$ is strictly decreasing. The firms independently and simultaneously choose outputs.

You are asked to prove the existence of a *symmetric* Nash equilibrium in the following steps. Let x denote the output of firm 1 and y the output of firm 2. Note that a firm's objective function need not be quasiconcave in its own output.

- (a) Argue that any output greater than $\bar{q}$ is a strictly dominated strategy for either firm. For the rest of the problem take each firm's strategy set to be $[0, \bar{q}]$.

- (b) Consider the problem for $y \in [0, \bar{q}]$:

$$\max_{z \in [y, \bar{q}]} (P(z) - c)(z - y).$$

In words, firm 1 chooses the *aggregate* output z to maximize its profit given $y \geq 0$ and subject to the additional constraint that $z \geq y$. Prove that if z_i solves this problem for $y = y_i$ for $i = 0, 1$ and $y_1 > y_0 \geq 0$, then $z_1 \geq z_0$.

(c) Return to the original formulation in which firm 1 chooses x . For $q \in [0, \bar{q}]$, let $r(q)$ be the *largest* best reply of a firm to its rival's output of q . (Since P is continuous and the domain of each firm's objective compact, a largest best reply exists for each $q \in [0, \bar{q}]$.) Show that if $y_1 > y_0 \geq 0$, then

$$r(y_1) - r(y_0) \geq -(y_1 - y_0)$$

The inequality implies that r cannot have "downward jumps." (A downward jump at y_1 would imply that the left side converges to a negative number as $y_0 \rightarrow y_1$ from the left.)

- (d) Let $x^* = \sup\{x \in [0, \bar{q}] \mid r(x) \geq x\}$. (Since $r(0) \geq 0$, the set in braces is not empty.) Prove that $x^* = r(x^*)$. I suggest you do so in the following sequence.

- i. Prove that $r(x^*) \geq x^*$. (Consider a sequence x_n of points in $\{x \in [0, \bar{q}] \mid r(x) \geq x\}$ with limit x^* and use the fact that $x_n \leq r(x_n)$ and $x_n \leq x^*$ for each n . If you cannot prove this part then *assume* it in what follows.)

- ii. Argue that $x^* < \bar{q}$.

- iii. Prove that $r(x^*) \leq x^*$. (Consider $x^* + \varepsilon < \bar{q}$ for $\varepsilon > 0$ and show that $\lim_{\varepsilon \rightarrow 0^+} r(x^* + \varepsilon) = r(x^*)$.)

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